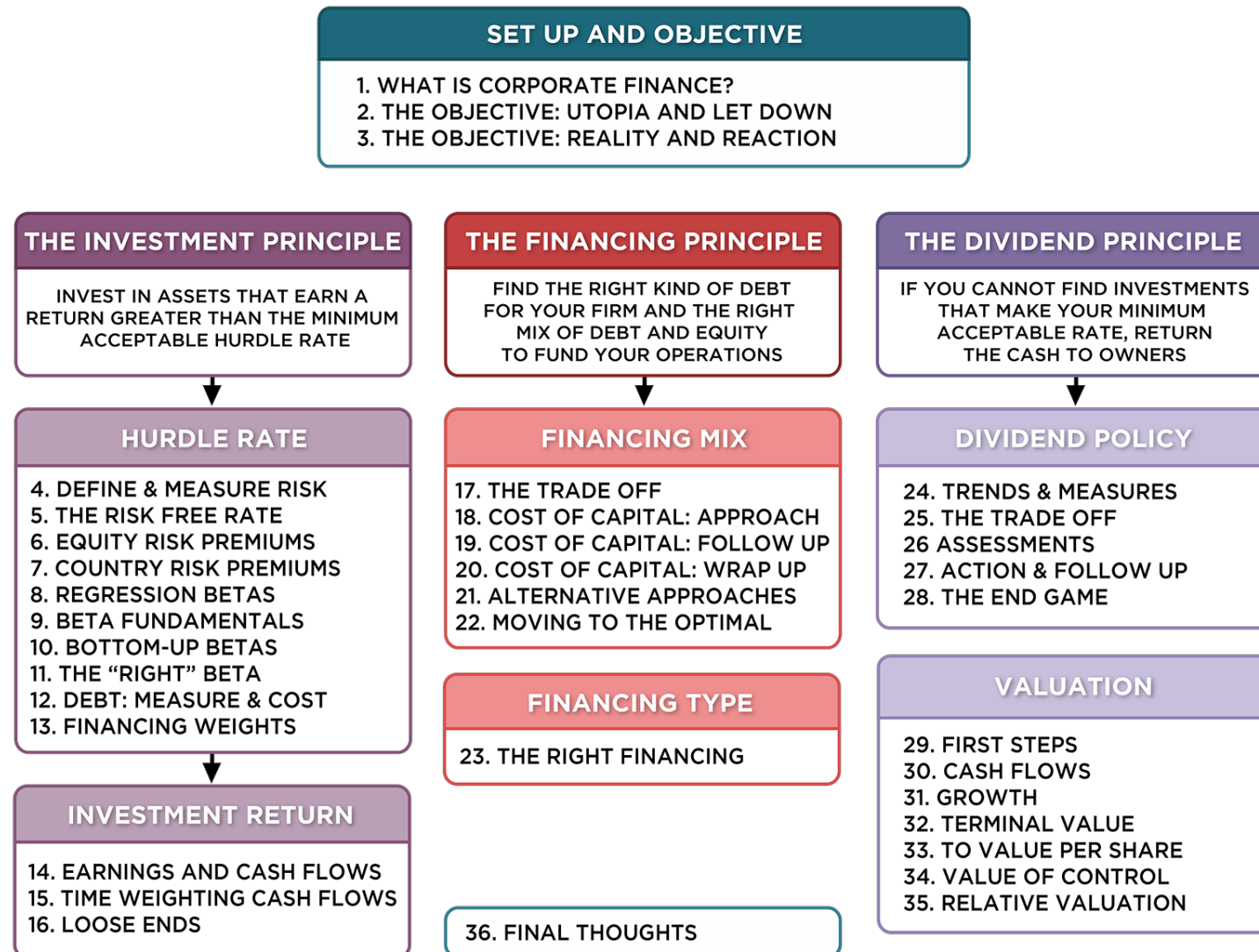


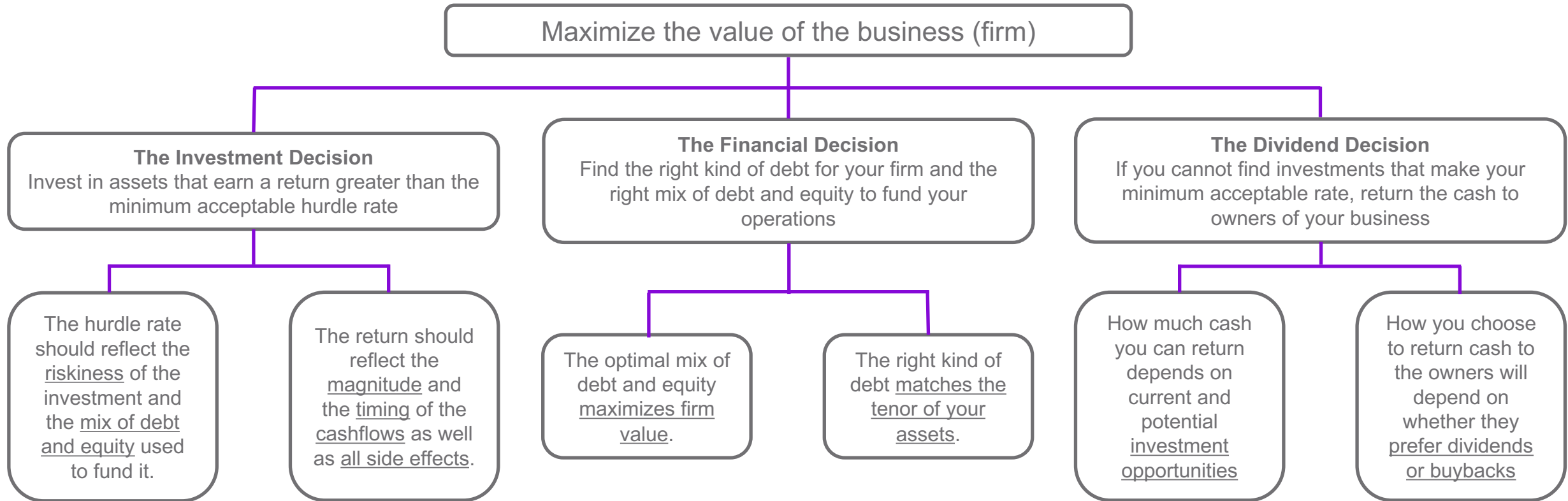
SESSION 14: EARNINGS AND CASH FLOWS

Show me the money

SESSION 14: EARNINGS AND CASH FLOWS



FIRST PRINCIPLES



MEASURES OF RETURN: EARNING VERSUS CASH FLOWS

- Principles governing Accounting Earning Management
 - **Accrual Accounting:** Show revenues when products and services are sold or provided, not when they are paid for. Show expenses associated with these revenues rather than cash expenses.
 - **Operating versus Capital Expenditure:** Only expenses associated with creating revenues in the current period should be treated as operating expenses. Expenses that create benefits over several periods are written off over multiple periods (as depreciation or amortization).
- To get from account earnings to cash flows:
 - you have to add back non-cash expenses (like depreciation)
 - you have to subtract out cash outflows which are not expensed (such as capital expenditures)
 - you have to make accrual revenues and expenses into cash revenues and expenses (by considering changes in the working capital).

MEASURING RETURNS RIGHT: THE BASIC PRINCIPLES

- Use cash flows rather than earnings. You cannot spend earnings.
- Use “incremental” cash flows relating to the investment decision, that is, cash flows that occur as a consequence of the decision, rather than total cash flows.
- Use “time-weighted” returns, that is, value cash flows that occur earlier more than cash flows that occur later.

The Return Mantra: “Time-weighted, Incremental Cash Flow Return”

EARNINGS VERSUS CASH FLOWS: A DISNEY THEME PARK

- The theme parks to be build near Rio, modeled on Euro Disney in Paris and Disney World in Orlando.
- The complex will include a “Magic Kingdom” to be constructed, beginning immediately, and becoming operational at the beginning of the second year, and a second theme park modeled on Epcot Center at Orlando to be constructed in the second and third year, and becoming operational at the beginning of the fourth year.
- The earning and cash flows are estimated in nominal U. S. Dollars.

KEY ASSUMPTIONS ON START UP AND CONSTRUCTION

- Disney has already spent \$0.5 billion researching the proposal and getting the necessary licenses for the park; none of this investment can be recovered if the park is not built. This expenditure has been capitalized and will be depreciated straight line over ten years to a salvage value of zero.
- Disney will face substantial construction costs, if it chooses to build the theme parks.
 - The cost of constructing Magic Kingdom will be \$3 billion, with \$2 billion to be spent right now, and \$1 billion to be spent one year from now.
 - The cost of constructing Epcot II will be \$1.5 billion, with \$1 billion to be spent at the end of the second year and \$0.5 billion at the end of the third year.
 - These investments will be depreciated based upon a depreciation schedule in the tax code, where depreciation will be different each year.

STEP 1: ESTIMATE ACCOUNTING EARNING ON PROJECT

	0	1	2	3	4	5	6	7	8	9	10
Magic Kingdom - Revenues		\$0	\$1,000	\$1,400	\$1,700	\$2,000	\$2,200	\$2,420	\$2,662	\$2,928	\$2,987
Epcot Rio - Revenues		\$0	\$0	\$0	\$300	\$500	\$550	\$605	\$666	\$732	\$747
Resort & Properties - Revenues		\$0	\$250	\$350	\$500	\$625	\$688	\$756	\$832	\$915	\$933
Total Revenues			\$1,250	\$1,750	\$2,500	\$3,125	\$3,438	\$3,781	\$4,159	\$4,575	\$4,667
Magic Kingdom – Direct Expenses		\$0	\$600	\$840	\$1,020	\$1,200	\$1,320	\$1,452	\$1,597	\$1,757	\$1,792
Epcot Rio – Direct Expenses		\$0	\$0	\$0	\$180	\$300	\$330	\$363	\$399	\$439	\$448
Resort & Property – Direct Expenses		\$0	\$188	\$263	\$375	\$469	\$516	\$567	\$624	\$686	\$700
Total Direct Expenses			\$788	\$1,103	\$1,575	\$1,969	\$2,166	\$2,382	\$2,620	\$2,882	\$2,940
Depreciation & Amortization		\$50	\$425	\$469	\$444	\$372	\$367	\$364	\$364	\$366	\$368
Allocated G&A Costs		\$0	\$188	\$263	\$375	\$469	\$516	\$567	\$624	\$686	\$700
Operating Income		-\$50	-\$150	-\$84	\$106	\$315	\$389	\$467	\$551	\$641	\$658
Taxes		-\$18	-\$54	-\$30	\$38	\$114	\$141	\$169	\$199	\$231	\$238
Operating Income after Taxes		-\$32	-\$96	-\$54	\$68	\$202	\$249	\$299	\$352	\$410	\$421

Direct expenses: 60% of revenues for theme parks, 75% of revenues for resort properties

Allocated G&A: Company G&A allocated to project, based on projected revenues. Two thirds of expense is fixed, rest is variable.

Taxes: Based on marginal tax rate of 36.1%

AND THE ACCOUNTING VIEW OF RETURN

Year	After-tax Operating Income	BV of pre-project investment	BV of fixed assets	BV of Working capital	BV of Capital	Average BV of Capital	ROC(a)	ROC(b)
0		500	2000	0	\$2,500			
1	-\$32	\$450	\$3,000	\$0	\$3,450	\$2,975	-1.07%	-1.28%
2	-\$96	\$400	\$3,813	\$63	\$4,275	\$3,863	-2.48%	-2.78%
3	-\$54	\$350	\$4,145	\$88	\$4,582	\$4,429	-1.22%	-1.26%
4	\$68	\$300	\$4,027	\$125	\$4,452	\$4,517	1.50%	1.48%
5	\$202	\$250	\$3,962	\$156	\$4,368	\$4,410	4.57%	4.53%
6	\$249	\$200	\$3,931	\$172	\$4,302	\$4,335	5.74%	5.69%
7	\$299	\$150	\$3,931	\$189	\$4,270	\$4,286	6.97%	6.94%
8	\$352	\$100	\$3,946	\$208	\$4,254	\$4,262	8.26%	8.24%
9	\$410	\$50	\$3,978	\$229	\$4,257	\$4,255	9.62%	9.63%
10	\$421	\$0	\$4,010	\$233	\$4,243	\$4,250	9.90%	9.89%
Average							4.18%	4.11%

- (a) Based upon book capital at the start of each year
(b) Based upon average book capital over the year

WHAT SHOULD THIS RETURN BE COMPARED TO?

- The computed return on capital on this investment is about 4%. TO make a judgment on whether this is a sufficient return, we need to compare this return to a “hurdle rate”. Which of the following is the right hurdle rate? Why or why not?
 - a. The riskfree rate of 2.75% (T-bond rate)
 - b. The cost of equity for Disney as a company (8.52%)
 - c. The cost of equity for Disney theme parks (7.09%)
 - d. The cost of capital for Disney as a company (7.81%)
 - e. The cost of capital for Disney theme parks (6.61%)
 - f. None of the above

SHOULD THERE BE A RISK PREMIUM FOR FOREIGN PROJECTS?

- The exchange rate risk should be diversifiable risk (and hence should not command a premium) if
 - the company has projects in a large number of countries, or
 - the investors in the company are globally diversified.
 - For Disney, this risk should not affect the cost of capital used. Consequently, we would not adjust the cost of capital for Disney's investments in other mature markets (Germany, U.K., France).
- The same diversification argument can also be applied against some political risk, which would mean that it too should not affect the discount rate. However, there are aspects of political risk, especially in emerging markets, that will be difficult to diversify and may affect the cash flows, by reducing the expected life or cash flows on the project.
- For Disney, this is the risk that we are incorporating into the cost of capital when it invests in Brazil (or any other emerging market).

ESTIMATING A HURDLE RATE FOR RIO DISNEY

- We did estimate a cost of capital of 6.61% for the Disney theme park business, using a bottom-up levered beta of 0.7537 for the business.
- This cost of equity market may not adequately reflect the additional risk associated with the theme park being in an emerging market.
- The only concern we would have with using this cost of equity for this project is that it may not adequately reflect the additional risk associated with the theme park being in an emerging market (Brazil). We first computed the Brazil country risk premium (by multiplying the default spread for Brazil by the relative equity market volatility) and then re-estimated the cost of equity:
 - Country risk premium for Brazil = 5.5% + 3% = 8.5%
 - Cost of Equity in US\$ = 2.75% + 0.7537 (8.5%) = 9.16%
- Using this estimate of the cost of equity, Disney's theme park debt ratio of 10.24% and its after-tax cost of debt of 2.40% (see chapter 4), we can estimate the cost of capital for the project:
 - Cost of Capital in US\$ = 9.16% (0.8976) + 2.40% (0.1024) = 8.46%

WOULD LEAD US TO CONCLUDE THAT . . .

- Do not invest in this park. The return on capital of 4.18% is lower than the cost of capital for theme parks of 8.46%. This would suggest that the project should not be taken.
- Given that we have computed the average over an arbitrary period of 10 years, while the theme park itself would have a life greater than 10 years, would you feel comfortable with this conclusion?
 - Yes
 - No

A TANGENT: FROM NEW TO EXISTING INVESTMENTS – ROC FOR THE ENTIRE FIRM

How “good” are the existing investments of the firm?

	ASSETS		LIABILITIES
EXISTING INVESTMENTS GENERATE CASH FLOWS TODAY INCLUDING LONG (FIXED) & SHORT (WORKING CAPITAL) ASSETS	ASSETS IN PLACE	DEBT	FIXED CLAIM ON CASH FLOWS LITTLE ROLE IN MANAGEMENT FIXED MATURITY TAX DEDUCTIBLE
EXPECTED VALUE THAT WILL BE CREATED BY FUTURE INVESTMENTS	GROWTH ASSETS	EQUITY	RESIDUAL CLAIM ON CASH FLOWS SIGNIFICANT ROLE IN MANAGEMENT PERPETUAL LIVES

MEASURING ROC FOR EXISTING INVESTMENTS								
Company	EBIT (1-t)	BV of Debt	BV of Equity	Cash	BV of Capital	Return on Capital	Cost of Capital	ROC – Cost of Capital
Disney	\$6,920	\$16,328	\$41,958	\$3,387	\$54,899	12.61%	7.81%	4.80%
Vale	\$12,432	\$49,246	\$75,974	\$5,818	\$119,402	10.41%	8.20%	2.22%
Baidu	¥9,111	¥13,561	¥27,215	¥10,456	¥30,320	30.05%	12.42%	17.63%
Tata Motors	₹120,905	₹471,489	₹330,056	₹225,562	₹575,983	20.99%	11.44%	9.55%
Bookscape	\$1,775	\$12,136	\$8,250	\$1,250	\$19,136	9.28%	10.30%	-1.02%

OLD WINE IN A NEW BOTTLE . . . ANOTHER WAY OF PRESENTING THE SAME RESULTS . . .

- The key to value is earning excess returns. Over time, there have been attempts to restate this obvious fact in new and different ways. For instance, Economic Value Added (EVA) developed a wide following in the 1990s:

$$\text{EVA} = (\text{ROC} - \text{Cost of Capital}) (\text{Book Value of Capital Invested})$$

Company	ROC – Cost of Capital	BV of Capital	EVA
Disney	4.80%	\$54,899	\$2,632
Vale	2.22%	\$119,402	\$2,645
Baidu	17.63%	\$30,320	\$5,347
Deutsche Bank	NMF	NMF	NMF
Tata Motors	9.55%	\$575,983	\$55,033
Bookscape	-1.02%	\$19,136	-\$195



APPLICATION TEST: ASSESSING INVESTMENT QUALITY

- For the most recent period for which you have data, compute the after-tax return on capital earned by your firm, where after-tax return on capital is computed to be:

$$\text{After-tax ROC} = \text{EBIT} (1 - \text{tax rate}) / (\text{BV of debt} + \text{BV of Equity} - \text{Cash})_{\text{previous year}}$$

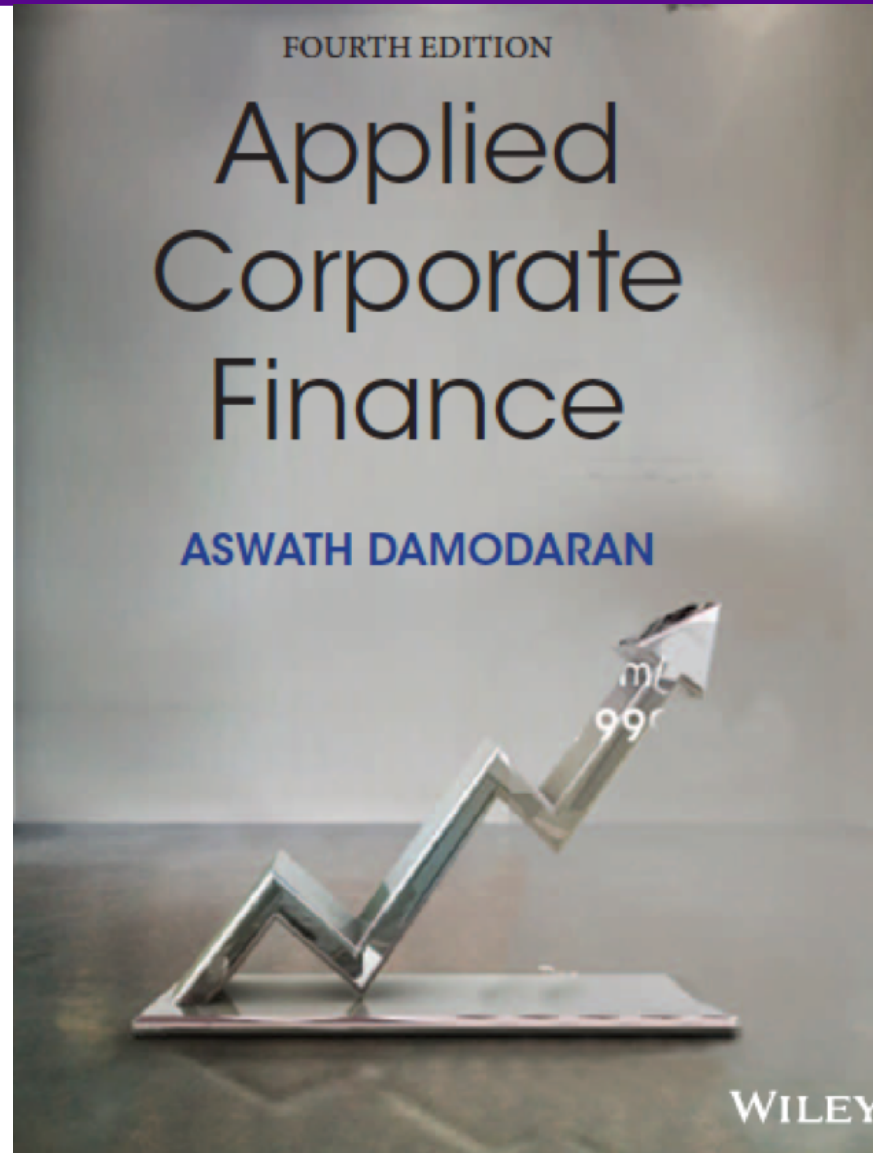
- For the most recent period for which you have data, compute the return spread earned by your firm:

$$\text{Return Spread} = \text{After-tax ROC} - \text{Cost of Capital}$$

- For the most recent period, compute the EVA earned by your firm:

$$\text{EVA} = \text{Return Spread} * ((\text{BV of Debt} + \text{BV of Equity} - \text{Cash})_{\text{previous year}})$$

Task
Estimate the
return
differential
(ROIC-WACC)
earned by your
company



Optional:
Read
Chapter 5,6